

# Static Gesture Classification — Run Report

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## Data paths

|                      |                                                                       |
|----------------------|-----------------------------------------------------------------------|
| Training data path 1 | /home/jestin/ThesisRepo/ML/NewTrainingData/NewNewDynamic · 720 trials |
| Total training paths | 1 · 720 trials total                                                  |
| External test root   | /home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/NewNewDynamic         |
| Report output dir    | /home/jestin/ThesisRepo/ML/NewReports/StaticReports/Right             |

## Sensor selection

|                |                                                                                                                                                                    |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hands          | left                                                                                                                                                               |
| Segments       | thumb, index, middle, ring, pinky, wrist                                                                                                                           |
| Modalities     | YPR, Accel, Flex                                                                                                                                                   |
| Sensor columns | 164 channels                                                                                                                                                       |
| Classes        | 6: Double_ClosedHand_Flip_OpenHand, Double_FingerTips, Double_Okay_Ring_Middle_Pinky, Double_OpenHand_Flip_ClosedHand, Double_OpenHand_LPunch_R_Punch, Double_Snap |

### Sensor grid (✓ enabled, ✗ disabled, — not present)

| Hand / Segment | IMU_mid | IMU_prox | Flex_MCP | Flex_PIP |
|----------------|---------|----------|----------|----------|
| left palm      | ✓       | —        | —        | —        |
| left thumb     | ✓       | ✓        | ✓        | ✓        |
| left index     | ✓       | ✓        | ✓        | ✓        |
| left middle    | ✓       | ✓        | ✓        | ✓        |
| left ring      | ✓       | ✓        | ✓        | ✓        |
| left pinky     | ✓       | ✓        | ✓        | ✓        |
| left wrist     | ✓       | —        | —        | —        |
| right palm     | ✓       | —        | —        | —        |
| right thumb    | ✓       | ✓        | ✓        | ✓        |
| right index    | ✓       | ✓        | ✓        | ✓        |
| right middle   | ✓       | ✓        | ✓        | ✓        |
| right ring     | ✓       | ✓        | ✓        | ✓        |
| right pinky    | ✓       | ✓        | ✓        | ✓        |
| right wrist    | ✓       | —        | —        | —        |

Enabled: 44 · Disabled: 0 (channels per enabled IMU depend on modality flags above).

## Preprocessing, features & split

|                             |                                            |
|-----------------------------|--------------------------------------------|
| Resample steps              | 90                                         |
| Butterworth                 | on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz |
| Normalisation               | minmax                                     |
| Feature mode                | stats+fft → feature dim 8856               |
| Test size · seed · CV folds | 0.2 · random_state=42 · 5-fold stratified  |
| Sample counts               | train (post-aug)=2880 · test=144           |

## Augmentation (training only)

|                        |                        |
|------------------------|------------------------|
| Enabled                | yes                    |
| Copies per train trial | 4                      |
| Jitter                 | off · sigma=0.01       |
| Amplitude scale        | on · range=(0.9, 1.1)  |
| Time warp              | off · range=(0.9, 1.1) |

## Classifiers — cross-validation & hold-out test

CV = 5-fold cross-validation. Test = hold-out accuracy. M-\* = macro-averaged, W-\* = support-weighted. Prec / Rec / F1 on hold-out test.

| Algorithm     | CV acc | CV std | CV F1  | CV F1 $\sigma$ | Test   | M-Prec | M-Rec  | M-F1   | W-Pre c | W-Rec  | W-F1   |
|---------------|--------|--------|--------|----------------|--------|--------|--------|--------|---------|--------|--------|
| Random Forest | 1.0000 | 0.0000 | 1.0000 | 0.0000         | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000  | 1.0000 | 1.0000 |

## Per-class metrics (hold-out test)

=== Random Forest ===

|                                 | precision | recall | f1-score | support |
|---------------------------------|-----------|--------|----------|---------|
| Double_ClosedHand_Flip_OpenHand | 1.00      | 1.00   | 1.00     | 24      |
| Double_FingerTips               | 1.00      | 1.00   | 1.00     | 24      |
| Double_Okay_Ring_Middle_Pinky   | 1.00      | 1.00   | 1.00     | 24      |
| Double_OpenHand_Flip_ClosedHand | 1.00      | 1.00   | 1.00     | 24      |
| Double_OpenHand_LPunch_R_Punch  | 1.00      | 1.00   | 1.00     | 24      |
| Double_Snap                     | 1.00      | 1.00   | 1.00     | 24      |
| accuracy                        |           |        | 1.00     | 144     |

|              |      |      |      |     |
|--------------|------|------|------|-----|
| macro avg    | 1.00 | 1.00 | 1.00 | 144 |
| weighted avg | 1.00 | 1.00 | 1.00 | 144 |

External test (NewTestData)

| Algorithm     | External test acc |
|---------------|-------------------|
| Random Forest | 0.8833            |

Per-class external accuracy

| Class                           | Random Forest | Support |
|---------------------------------|---------------|---------|
| Double_ClosedHand_Flip_OpenHand | 1.000         | 20      |
| Double_FingerTips               | 0.300         | 20      |
| Double_Okay_Ring_Middle_Pinky   | 1.000         | 20      |
| Double_OpenHand_Flip_ClosedHand | 1.000         | 20      |
| Double_OpenHand_LPunch_R_Punch  | 1.000         | 20      |
| Double_Snap                     | 1.000         | 20      |